

Hands On:
La Plata River Basin Example

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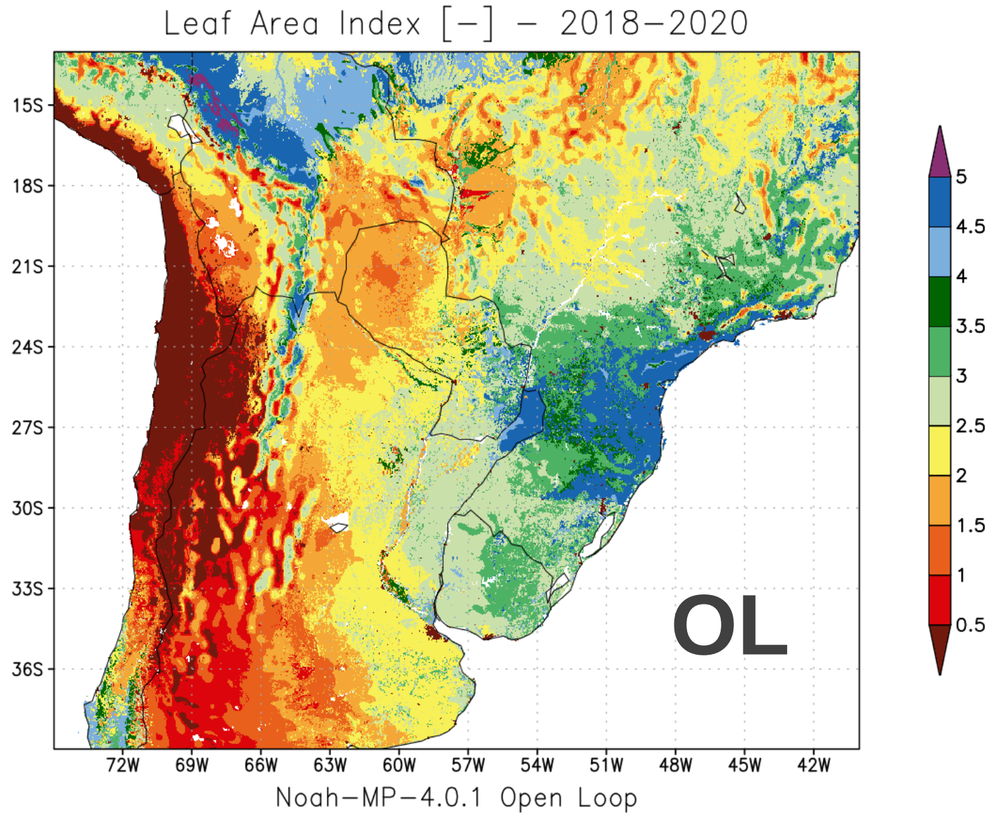
LDT/LIS specifications

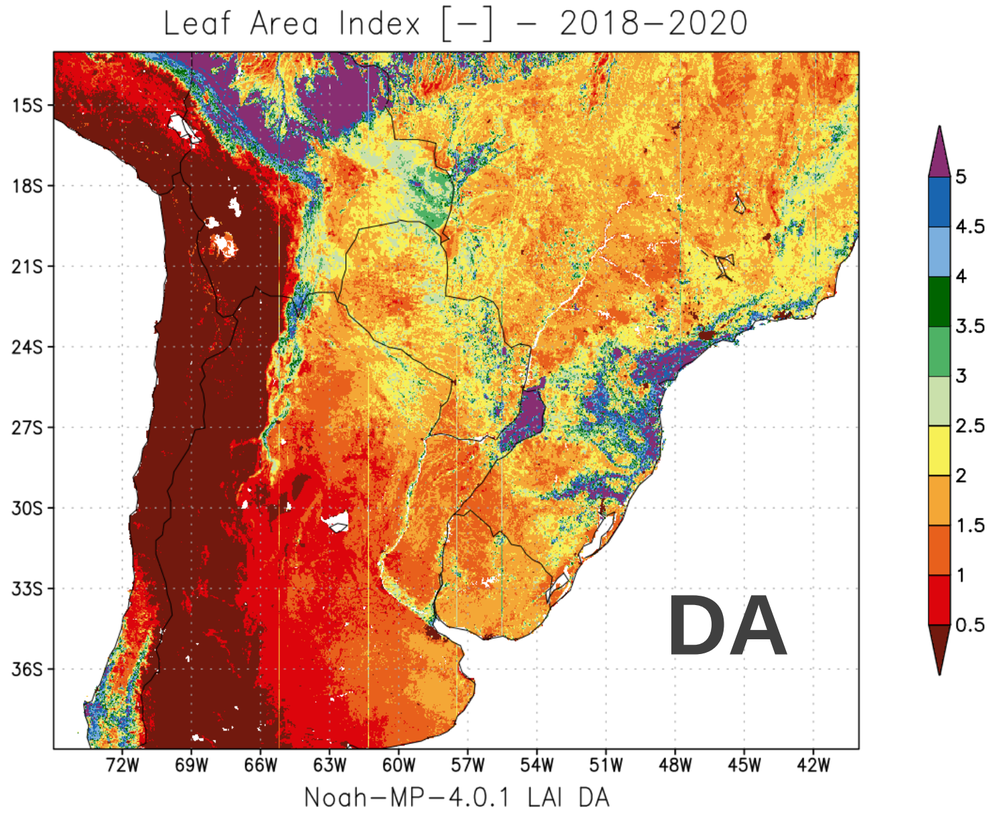
- We first ran LDT to set a domain of 0.04-deg. (4-km) from 39-13 South & 75-40 West
- We chose the "**MODIS IGBP**" landcover classification and **ISRIC** soil textures
- We chose the Noah-MP-4.0.1 land-surface model for the simulations
  - We selected the "prognostic phenology" vegetation physics option in Noah-MP
- We chose the GDAS surface meteorological dataset for the forcing
- Two different streamflow routing schemes are available:
  - HyMAP or RAPID
- Two sets of LIS simulations are available for each routing scheme:
  - Open Loop (OL) – the Noah-MP LSM prognostically simulates the LAI
  - Data Assimilation (DA) – MODIS LAI is assimilated into the Noah-MP LSM

LIS simulations

- In advance of this week's training, we performed a "spin-up" of the model states to approach equilibrium. We performed 57 years of simulation, up to 1 Jan 2018.
- We then ran 3 years of simulation from 1 Jan 2018 to 1 Jan 2021, for a total of four different experiments (HyMAP or RAPID – and – OL or DA).
- We used the MODIS MCD15A2H v006 LAI dataset for the assimilation.
- These 3 year "canned" experiments with daily output are available on your server.
- On the next few slides, we will take a quick look at some of the simulated results from the 3-year "canned" simulations.
- HOWEVER, for this week's training, we will instead try running some "use case" experiments, which are just 3-day simulations, to allow the simulations to finish running on your server in a timely manner.

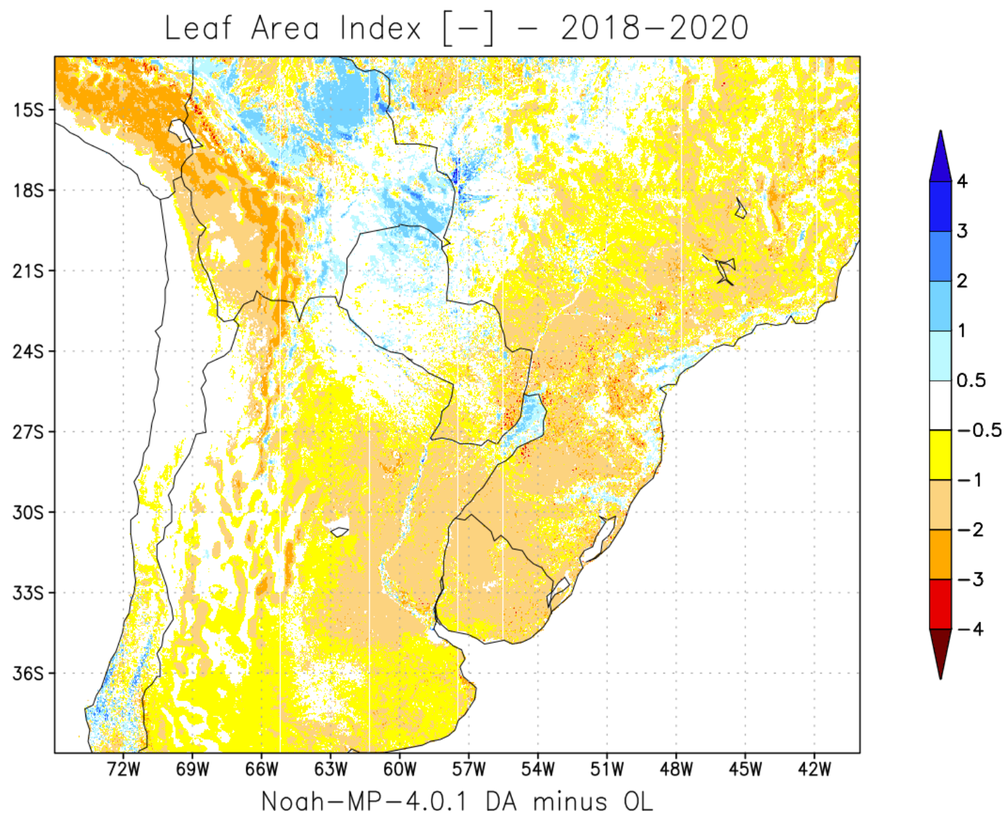
Simulated LAI from OL and DA

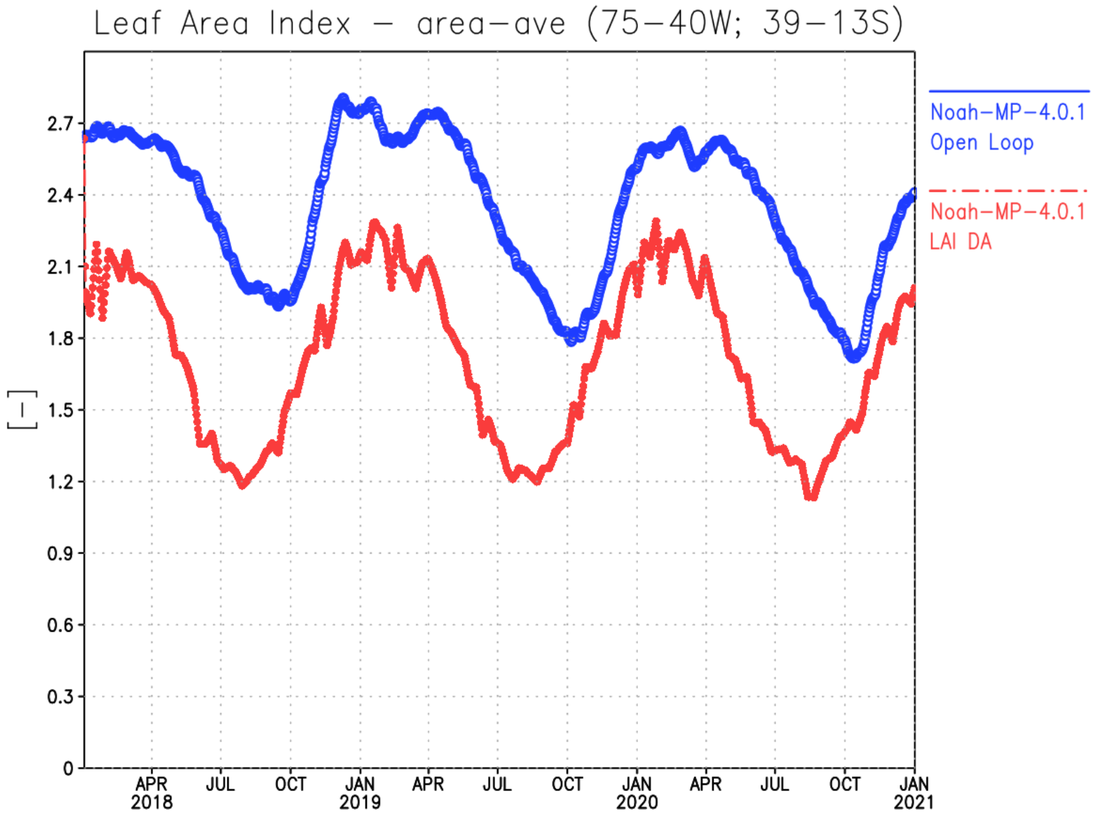




- 3-year average of LAI from OL (left) and from DA (right)
- The DA simulation generally has lower LAI over much of the domain, while the OL simulation has lower LAI in the northeastern portion of Argentina

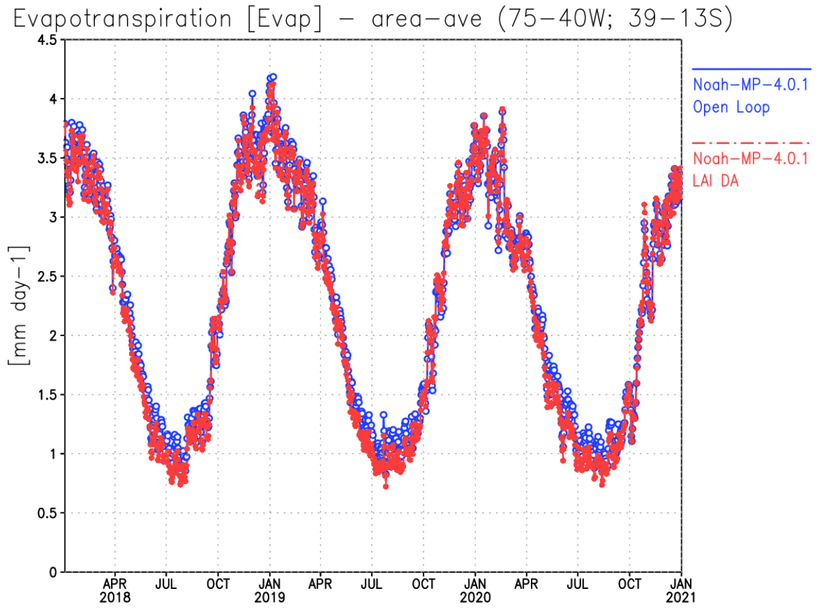
Simulated LAI from OL and DA

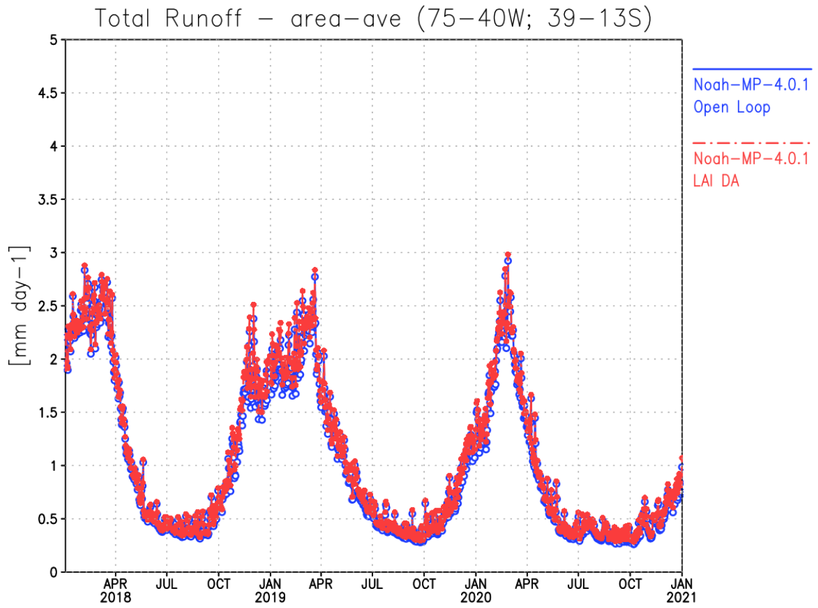


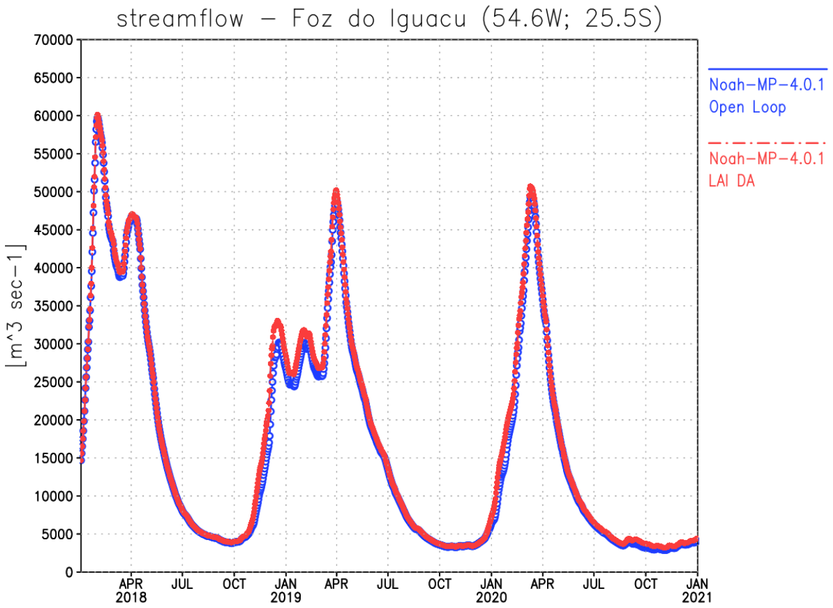


- 3-year LAI difference (DA minus OL; left) shows mostly lower LAI over the domain
- Entire domain area-average time series (right) of LAI shows the seasonal cycle, with lower LAI in the DA simulation

Simulated Evapotranspiration, Runoff, Streamflow







- (Left) Evapotranspiration is slightly lower in the DA simulation, from the lower LAI
- (Middle) The simulated total runoff is slightly higher in the DA simulation
- (Right) Routing the runoff into streamflow using the HyMAP router shows that the streamflow at Foz do Iguaçu (54.6W & 25.5S) is higher in the DA simulation, notably from ~Nov 2018 to ~Apr 2019, where the upstream LAI is generally lower

Looking at a “use case” lis.config file

```
#Overall driver options
Running mode:                                   "retrospective"
Map projection of the LIS domain:               latlon
Number of nests:                                1
Number of surface model types:                  2
Surface model types:                            "LSM"  "Openwater"
Surface model output interval:                  1da
Land surface model:                             "Noah-MP.4.0.1"
Open water model:                               "template open water"
Number of met forcing sources:                  1
Blending method for forcings:                   overlay
Met forcing sources:                            "GDAS"
Met forcing chosen ensemble member:             1
Topographic correction method (met forcing):  "lapse-rate and slope-aspect"
```

- “retrospective” running mode is for historical simulation
- 2 ”surface model” types are used: 1 for the land-surface model (Noah-MP-4.0.1 LSM) and 1 for “openwater” points. The openwater allows for the streamflow routers to simulate the water flow for non-land grid points. Daily output.
- We will use the “GDAS” surface meteorological forcing dataset to drive the LSM.

- The surface meteorology is corrected to our grid using the topography (elevation data) at each grid.
- Temperature/humidity/pressure are corrected using a standard lapse-rate difference between the height of GDAS and the height of the grid point (MERIT).
- Shortwave down (solar) corrected by slope-aspect.

Looking at a “use case” lis.config file

```
Start mode:                    restart
Starting year:                 2019
Starting month:                   3
Starting day:                    31
Starting hour:                   23
Starting minute:                 45
Starting second:                  0
Ending year:                   2019
Ending month:                     4
Ending day:                       4
Ending hour:                      0
Ending minute:                    0
Ending second:                    0
```

- We will use a “restart” file for model initialization. We have provided this file from our long model spin-up, and represents the model states at this time.
- The “valid” time of the restart file is at 23:45Z on 31 March 2019. For our spin-up, we chose to write a restart file at the end of every month.
- We are running a 3-day LIS simulation, from ~1 April 2019 to 4 April 2019.

Looking at a “use case” lis.config file

```
#------------------------DOMAIN SPECIFICATION--------------------------
#The following options list the choice of parameter maps to be used
LIS domain and parameter data file:      ./lis_input.laplata.noahmp401.4km.nc
Landmask data source:                    LDT
Landcover data source:                   LDT
Soil texture data source:                LDT
Soil fraction data source:               none
Soil color data source:                  none
Elevation data source:                   LDT
Slope data source:                       LDT
Aspect data source:                      LDT
Curvature data source:                   none
LAI data source:                         none
SAI data source:                         none
Albedo data source:                      none
Max snow albedo data source:             LDT
Greenness data source:                   LDT
```

- Our simulation domain and parameters were set in LDT, which created the “lis_input” file which LIS will use.
- “LDT” indicates that LIS will use these parameters as they were set in the “lis_input” file created by LDT.

Looking at a “use case” lis.config file

```
#------------------------------FORCINGS-------------------------------------
GDAS forcing directory:                          ./input/MET_FORCING/GDAS

#----------------------LAND SURFACE MODELS--------------------------
Noah-MP.4.0.1 model timestep:                    15mn
Noah-MP.4.0.1 restart output interval:           1mo
Noah-MP.4.0.1 restart file:                      ./LIS_RST_NOAHMP401_201903312345.d01.nc
Noah-MP.4.0.1 restart file format:               netcdf
Noah-MP.4.0.1 soil parameter table:              ./input/LS_PARAMETERS/noahmp401_parms/SOILPARM.TBL
Noah-MP.4.0.1 general parameter table:           ./input/LS_PARAMETERS/noahmp401_parms/GENPARM.TBL
Noah-MP.4.0.1 MP parameter table:                ./input/LS_PARAMETERS/noahmp401_parms/MPTABLE.TBL
Noah-MP.4.0.1 number of soil layers:             4
Noah-MP.4.0.1 thickness of soil layers:          0.1  0.3  0.6  1.0
```

- Point to the directory with the GDAS surface meteorological forcing
- The Noah-MP-4.0.1 LSM will run with a 15-minute timestep. Point to the parameter tables, and use four soil layers (0-10cm, 10-40cm, 40-100cm, and 100-200cm).
- Use the Noah-MP restart file valid at 23:45Z on 31 March 2019.

Looking at a “use case” lis.config file

```
#Sub-models
Routing model:                          "RAPID router"
Radiative transfer model:               none
--
#---------------------------ROUTING-----------------------------------
RAPID river routing time step:                15mn
RAPID routing model time step:                1da
RAPID routing model output interval:          1da
RAPID routing model restart interval:         1mo
RAPID routing model start mode:               restart
RAPID routing model restart file:             ./LIS_RST_RAPID_router_201903312345.d01.nc
RAPID run in ensemble mode:                   0 # 0=open loop; 1=ensemble mean; 2(not support)=ensemble

RAPID run option:                             1 # 1=regular run
RAPID routing option:                         1 # 1=matrix-based Muskingum; 2=traditional Muskingum; 3=
skingum operator
RAPID cost function phi option:               1 # 1=phi1; 2=phi2

RAPID initial flow:                           .false.
RAPID write final flow:                       .false.
RAPID compute volume:                         .false.
RAPID human-induced flow:                     .false.
RAPID upstream forcing:                       .false.
RAPID dam model used:                         .false.
RAPID output influence:                       .false.
RAPID uncertainty quantification:             .false.

RAPID namelist file:                          ./rapid_params/rapid_namelist
RAPID river connectivity file:                ./rapid_params/rapid_connect.csv
RAPID river weight table:                     ./rapid_params/weight_lis.csv
RAPID river basin ID file:                    ./rapid_params/riv_bas_id.csv
RAPID Muskingum parameter k file:             ./rapid_params/k.csv
RAPID Muskingum parameter x file:             ./rapid_params/x.csv
RAPID max number of upstream reaches:         3
```

- Using the RAPID streamflow router
- RAPID also runs with a 15-minute timestep, and daily output.
- Use a RAPID restart file, also valid at 23:45Z on 31 March 2019.
- Set RAPID physics options
- Point to RAPID parameter files

Looking at a “use case” lis.config file

```
Model output attributes file:                ./tables/NOAHMP401_OUTPUT_LIST.TBL.dyn.full
```

- We select which variables we want to output from the simulation in the “Model output attributes file”.
- A sample of this file is shown below:

```
#Surface state variables
SnowT:          0  K        -    1 0 0 1 165 10      # Snow surface temperature (K)
VegT:           1  K        -    1 0 0 1 146 10      # Vegetation canopy temperature (K)
BareSoilT:      1  K        -    1 0 0 1 147 10      # Temperature of bare soil (K)
AvgSurfT:       1  K        -    1 0 0 1 148 10      # Average surface temperature (K)
RadT:           1  K        -    1 0 0 1 149 10      # Surface radiative temperature (K)
Albedo:         1  -        -    1 0 0 1  84 100     # Surface albedo (-)
SWE:            1  kg/m2    -    1 0 0 1  65 1000    # Snow Water Equivalent (kg/m2)
SWEVeg:         0  kg/m2    -    1 0 0 1 139 1000    # SWE intercepted by vegetation (kg/m2)
SurfStor:       0  kg/m2    -    1 0 0 1 150 1000    # Surface water storage (kg/m2)

#Subsurface state variables
SoilMoist:      1  m3/m3    -    1 0 0 4  86 1000    # Average layer soil moisture (kg/m2)
SoilTemp:       1  K        -    1 0 0 4  85 1000    # Average layer soil temperature (K)
SmLiqFrac:      0  -        -    1 0 0 4 160 100     # Average layer fraction of liquid moisture (-)
SmFrozFrac:     0  -        -    1 0 0 4 140 100     # Average layer fraction of frozen moisture (-)
SoilWet:        0  -        -    1 0 0 1 144 100     # Total soil wetness (-)
RelSMC:         0  m3/m3    -    1 0 0 4 141 1000    # Relative soil moisture (-)
RootTemp:       0  K        -    1 0 0 1 142 1000    # Root zone temperature (K)
```

- Value in the 2nd column determines if this variable is written or not (0 = no; 1 = yes)
- Units of the output are set in the 3rd column
- 4th column determines flux direction (or none)
- Further details of the other column options are in the LIS Users’ Guide documentation

Looking at the “use case” files/directories

```
-rw-r--r-- 1 dmocko s1189 36322511 Jan 22 23:28 LIS_RST_NOAHMP401_201903312345.d01.nc
-rw-r--r-- 1 dmocko s1189  2693389 Jan 22 23:28 LIS_RST_RAPID_router_201903312345.d01.nc
-rw-r--r-- 1 dmocko s1189     1547 May 28  2014 forcing_variables.txt
drwxr-xr-x 4 dmocko s1189     4096 Apr 19  2019 input/
-rw-r--r-- 1 dmocko s1189    12251 Feb 21 10:49 lis.config.laplata.noahmp401.rapid.4km.OL.use-case
-rw-r--r-- 1 dmocko s1189 57238223 Aug 29  2023 lis_input.laplata.noahmp401.4km.nc
drwxr-xr-x 2 dmocko s1189     4096 Aug 18  2023 rapid_params/
drwxr-xr-x 2 dmocko s1189     4096 Sep 14 11:11 tables/
drwxr-xr-x 5 dmocko s1189     8192 Feb 21 11:16 target_OL-use_case/
```

- To run our “use case”, we have provided you with the above files/directories:
  1) Restart files for Noah-MP-4.0.1 LSM and for RAPID: “LIS_RST”
  2) “forcing_variable.txt” – list of GDAS forcing variables and their units
  3) “input” – directory with GDAS forcing and Noah-MP-4.0.1 parameter table files
  4) lis.config file to run this use case
  5) “lis_input” file for this use case (generated by previously running LDT)
  6) “rapid_params” – directory with RAPID routing model parameter files
  7) “tables” – directory with the Model Output attributes file for this simulation

Looking at the “use case” files/directories

```
-rw-r--r-- 1 dmocko s1189 36322511 Jan 22 23:28 LIS_RST_NOAHMP401_201903312345.d01.nc
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drwxr-xr-x 2 dmocko s1189     4096 Sep 14 11:11 tables/
drwxr-xr-x 5 dmocko s1189     8192 Feb 21 11:16 target_OL-use_case/
```

- We have already run this “use case” for you, and the output for it is in the directory “target_OL-use-case”
- We have provided this output to you, so we can compare your “live” running of this use case against the “target”
- If the outputs are similar, then we know that you have successfully run this “use case”
- Note that there may be from very small insignificant differences in the outputs, as they were run on different computers with different code compilers, etc.

Looking at the “use case” stats output

- As the model run, it will generate an ASCII/text file: SURFACEMODEL.d01.stats

Statistical Summary of LSM output for: 4/ 2/2019 0: 0: 0

	Mean	Stdev	Min	Max
lat(degree_north)	-25.995	7.505	-38.980	-13.020
lon(degree_east)	-57.493	10.103	-74.980	-40.020
Swnet(W m-2)	190.959	39.717	5.816	298.596
Lwnet(W m-2)	-66.606	26.632	-172.865	21.979
Qle(W m-2)	84.471	45.133	-6.757	199.610
Qh(W m-2)	37.252	27.233	-49.103	136.742
Qg(W m-2)	2.473	5.774	-67.249	92.240
Snowf(kg m-2 s-1)	0.868E-07	0.358E-05	0.000E+00	0.688E-03
Rainf(kg m-2 s-1)	0.537E-04	0.108E-03	0.000E+00	0.128E-02
Evap(kg m-2 s-1)	0.337E-04	0.180E-04	-0.238E-05	0.795E-04
Qs(kg m-2 s-1)	0.515E-05	0.188E-04	0.000E+00	0.596E-03
Qsb(kg m-2 s-1)	0.133E-04	0.234E-04	-0.721E-05	0.691E-03
Qsm(kg m-2 s-1)	0.242E-06	0.852E-05	0.000E+00	0.795E-03
VegT(K)	294.749	6.691	251.079	307.159
BareSoilT(K)	296.060	5.401	250.355	312.250
AvgSurfT(K)	295.845	5.341	250.355	312.250
RadT(K)	296.038	5.344	250.355	312.250
Albedo(-)	0.171	0.043	0.088	0.840
SWE(kg m-2)	2.156	62.140	0.000	2000.001
SnowDepth(m)	0.004	0.130	0.000	5.489
SoilMoist(m^3 m-3)	0.228	0.082	0.017	1.000
SoilMoist(m^3 m-3)	0.245	0.061	0.048	1.000
SoilMoist(m^3 m-3)	0.238	0.070	0.043	1.000
SoilMoist(m^3 m-3)	0.257	0.066	0.047	1.000
SoilTemp(K)	296.650	5.344	258.903	312.211
SoilTemp(K)	296.186	5.180	258.872	311.957
SoilTemp(K)	295.418	5.174	258.822	310.861
SoilTemp(K)	295.281	5.356	258.733	309.160

- This file provides some domain-average values for each variable
- The time of the output is shown at the top: 00Z on 2 April 2019
- For each variable there are 4 columns:
  1. Mean (average of entire domain)
  2. Stdev (standard deviation within the entire domain)
  3. Min (minimum value in the domain)
  4. Max (maximum value in the domain)

Running the “use case”

- To run the “use case”, on the command line, type this command:

```
mpirun -np 36 ./LIS -f lis.config.laplata.noahmp401.rapid.4km.OL.use-case
```

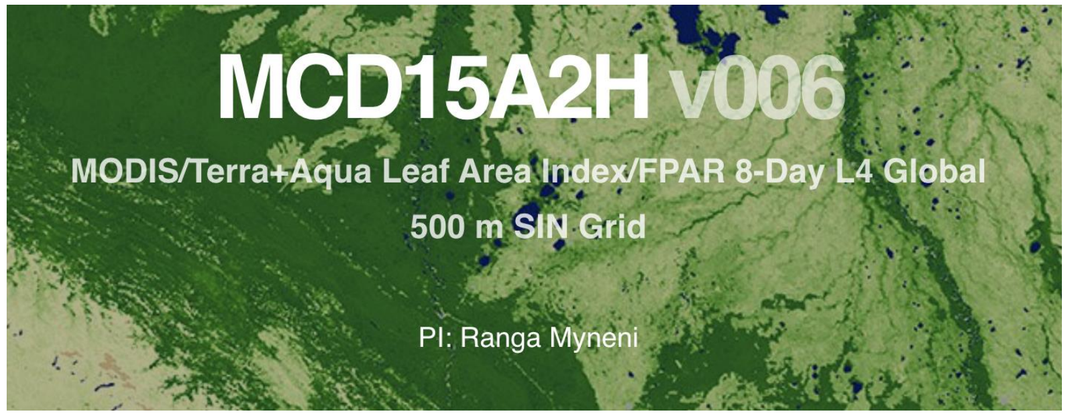
- We are telling our machine to run in parallel, using 36 different processors
- The lis.config file specifies that we are running a 6 by 6 processor layout
- The `-f` is needed to tell LIS to use this particular lis.config file
- We expect that the simulation will need about 10-12 minutes to complete

Other “use case” packages available

- The first “use case” we are trying is for the RAPID router with “Open Loop”
- We have also provided a “use case” for RAPID with LAI data assimilation, as well as for the HyMAP router both OL and DA
- These ”use case”s include the needed lis.config files, parameter datasets, restarts, etc.
- The LAI data assimilation lis.config files include:

```
#-------------------DATA ASSIMILATION ------------------------------
#Data assimilation options
Number of data assimilation instances:                    1
Data assimilation algorithm:                              "EnKF"
Data assimilation set:                                    "MCD15A2H LAI"

MCD15A2H LAI data directory:                              ./input/LS_PARAMETERS/MODIS/MCD15A2H.006
MCD15A2H LAI data version:                                "006"
MCD15A2H LAI apply temporal smoother between 8-day intervals: 1
MCD15A2H LAI apply climatological fill values:            1
MCD15A2H LAI apply QC flags:                              1
```



- We are applying the following options for our run: a) temporal smoother between the 8-day data; b) fill any missing values with climatology; c) apply the LAI QC flags